

Scalability & Future Plan

Date	05 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The ML regression model is trained once on a static historical dataset and does not automatically sync with new socio-economic data releases.	Predictions may gradually lose alignment with current global developmental trends as UNDP criteria shift.	High
2	The interface only supports single-country simulations with no comparative tables or country rankings.	Limits the depth of decision support for development planners comparing multiple regional tracks.	Medium
3	No live database is used; the application relies on loading flat CSV and serialized .pkl files in-memory.	Historical target queries and user inputs are not saved, preventing trend analysis and user customization.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Deployed as a single Flask application; suitable for low-to-moderate traffic.	Implement containerized deployment (Docker) on auto-scaling instances with an API caching layer (Redis) for recurrent queries.
2	Data Storage	Flat-file caching of the indicators database (HDI.csv); no user query histories are saved.	Integrate a relational cloud database (PostgreSQL) to store simulation inputs and histories, enabling analytics.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	Performance	Serialized model files loaded from disk into memory, causing cold start delays.	Adopt container keep-warm services (scheduled pings) and optimize inference using lightweight formats (ONNX).
4	Security	Basic public routing with client-side form constraints; no authentication.	Add server-side boundary checks, request rate-limiting, and token-based admin access (JWT) for database management.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Multi-country simulation comparison grids and policy target ranking based on health/education budgets.	Q4 2026	Enables side-by-side comparative analysis of policy proposals.
Phase 3	Highly optimized mobile layout with offline data-entry support.	Q1 2027	Expands usability for field research teams in areas with limited internet access.
Phase 4	Automated integrations with live UNDP data endpoints.	Q3 2027	Ensures baseline country indicators are automatically updated without manual file changes.